MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, ETCHED FOIL), 85°C (VOLTAGE DERATED TO 125°C),
ESTABLISHED RELIABILITY,
STYLE CLR25

This amendment forms a part of Military Specification Sheet
MIL-C-39006/1C, dated 10 January 1972, and is approved for use
by all Departments and Agencies of the Department of Defense.

PAGE 1

* Following preamble, add:

*
CAUTIONARY NOTE: THESE CAPACITORS MAY EXPLODE IF SUBJECTED TO REVERSE VOLTAGES. WHEN FIRST TESTED FOR DC LEAKAGE OR AT OTHER HIGH POTENTIALS, EXERCISE CAUTION TO VERIFY CORRECT POLARITY MARKINGS.

*
FAILURE RATE LEVEL 'L' INACTIVE FOR DESIGN AFTER 27 May 1976

*
CASE SIZE E1 INACTIVE FOR DESIGN AFTER 20 Oct 1978

FIGURE 1 (dimension table): Add G1 case size

G1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)	

PAGE 2

TABLE I, following part no. M39006/01-: Add "1/".

TABLE I, part number column: Add "-" following numbers 3030 through 3145 to denote optional random vibration part numbers.

TABLE I: Delete line 15 in its entirety. For replacement use part numbers M39006/01-3200, -3201-, -3203-, -3204-, and -3205-.

TABLE I, following line 15: Add new item:

50	75.0	+50, -15	4	70	30	57.5	52	G3	3200	3201-	3203-	3204-	3205-

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:14Z
Check the source to verify that this is the current version before use.

TABLE I: Add the new G1 case size units.

Rated voltage (85°C)	Cap.	Cap. tolerance	DC leakage (max) 25°C	DC leakage (max) 85°C 125°C	De-rated voltage (125°C)	Surge voltage (85°C)	Impedance (max)	Case size	Part no. 139006/01- 1/ Failure rate level (%/1,000 hr) L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
15	15.0	+75,-15	2	4	10	17.2	220	G1	3300	3307-	3314-	3321-	3328-
25	10.0	+75,-15	2	4	15	28.8	330	G1	3301	3308-	3315-	3322-	3329-
30	8.0	+75,-15	2	6	20	34.5	424	G1	3302	3309-	3316-	3323-	3330-
50	4.5	+50,-15	2	4	30	57.5	730	G1	3303	3310-	3317-	3324-	3331-
75	3.0	+50,-15	2	4	50	86.2	1000	G1	3304	3311-	3318-	3325-	3332-
100	2.0	+30,-15	2	4	65	115.0	1600	G1	3305	3312-	3319-	3326-	3333-
150	1.0	+30,-15	2	4	100	172.0	3000	G1	3306	3313-	3320-	3327-	3334-

Following Table I: Add "1/ Dash number for failure rate level M, P, R, and S parts will include the letter 'H' to indicate the optional random vibration requirement of '-' will be deleted."

PAGE 3

Following vibration requirement, add new Random vibration requirement:

"Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-E (18.7 g)."

Temperature cycling: Delete "Temperature cycling: Method 102 of MIL-STD-202, condition D" and substitute "Thermal shock: Method 107 of MIL-STD-202, condition A."

PAGE 4

Life at 85°C: Delete "240-hr (group B)" requirement in its entirety; also delete "10,000-hr (group C)" and substitute "10,000-hr (group B)".

Add the following new requirement under "Life at 125°C":

"10,000-hr (group B):

DCL (125°C) - 125% (max) of table I value.

DCL (25°C) - See table I.

ΔCap. - Within ±50% of initial measurement for E1 and G1 cases; within ±25% of initial measurement for all other cases.

DF - 80% (≤30 V) or 60% (>30 V) for E1 and G1 cases; 40% (≤30 V) or 30% (>30 V) (max) for all other cases.

DWV - 2,000 V dc, min.

IR - 100 megohms, min."

Marking: Delete requirement in its entirety and substitute:

"Marking: In accordance with MIL-C-39006.

Polarity - Series of plus (+) symbols encircling positive end (see figure 1)."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:14Z
Check the source to verify that this is the current version before use.

NOTE: The margin of this amendment is marked with an asterisk to indicate where a change (additions, modifications, corrections, deletions) from the previous amendment was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1535-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:14Z
Check the source to verify that this is the current version before use.